

Stockist Funds

Systematic funds, finance sentiment and an inspectable investment app

Project B — Data Factory Floor Stations 3 and 4

FINS5545 Financial Market Data Design & Analysis

z5577874

14 August 2026

Historical sample: 1 January 2020 to 31 December 2023

Deployed URL: <https://z5577874projectb.streamlit.app/?view=Overview>

Repository URL: https://github.com/Worryinglamb08/z5577874_projectB

Abstract

Stockist Funds turns the supplied 2020–2023 equity, cryptocurrency and headline data into 15 monthly systematic funds and an investor-facing app. The walk-forward evidence does not identify one dominant method. Combined Risk Parity records the strongest combined-fund Sharpe ratio (0.883), with 13.9% annualised return, 16.2% volatility and a 19.5% maximum drawdown. Combined Equal Weight returns 15.0% but draws down 27.9%. Crypto Minimum Variance returns 69.5%, yet loses 72.9% from peak to trough, so its return cannot be read apart from its risk. A finance-adjusted VADER lexicon changes 9,481 headline scores and lowers the neutral share by 2.90 percentage points. Coverage confidence ranges from 0.39 for Materials to 0.82 for Consumer. The fixed one-day-lagged coverage-aware tilt does not improve Equity Minimum Variance: Sharpe falls from 0.504 to 0.453 while maximum drawdown becomes 0.77 percentage points shallower. The app therefore places risk, holdings, coverage and limitations beside return instead of ranking funds. The results support a cautious product interpretation: systematic rules can be inspectable without treating one historical optimum as a forecast.

1. Product decision and scope

Stockist Funds is a historical prototype for a self-directed investor who wants a repeatable allocation rule but still wants to inspect what the rule did. The product offers equity, cryptocurrency and combined funds. Each fund pairs one asset family with one portfolio method, so the investor chooses a defined object rather than an unlabeled optimiser. The app supports fund comparison, a dated fact sheet, a hypothetical allocation and a news-signal view. It does not accept money, place trades or give personal advice at this stage.

The design responds to a simple tension. A managed fund can make a diversified portfolio easier to access, but its value can fall and its suitability depends on time horizon, tolerance for loss, liquidity and fees (Australian Securities and Investments Commission, 2026). A return leaderboard would hide that tension. Stockist Funds instead keeps annualised return, volatility, Sharpe ratio and maximum drawdown together; holdings, turnover and costs sit one layer below. The product language says “historical out-of-sample simulation” and dates every target weight.

The analysis compares five allocation methods across equity, cryptocurrency and combined funds, including Hierarchical Risk Parity (HRP). The headline analysis separates sentiment from the amount and concentration of news supporting it through a coverage-confidence measure. These results are presented in a Streamlit app that reads precomputed files for interactive exploration. The complete fund comparison appears in Table 2, while the method and family patterns appear in Figure 1 and Figure 3.

2. Data and walk-forward design

2.1 Reconciled inputs

The analysis builds on the cleaned Project A dataset so that Part B uses the same reconciled sample rather than creating a separate data foundation. The cleaned inputs contain 50,300 equity price rows for 50 companies, 14,610 cryptocurrency rows for ten assets and 146,836 unique ticker-date-title headlines. The 2,847 removed news rows are exact duplicates. Ten crypto observations after 31 December 2023 are also excluded. Seventeen reconciliation checks match the frozen Project A results exactly.

Equity returns use observed US trading days and 252-day annualisation, while Crypto-only funds retain their seven-day calendar and use 365 days. I retained the equity calendar for Combined funds and left-aligned crypto returns to those dates without carrying prices or returns forward. This choice avoids inventing a weekend equity return and keeps the combined investment decision on a calendar that all 50 stocks can follow.

2.2 Fifteen monthly funds

The menu contains Equal Weight, Minimum Variance, Risk Parity, Maximum Sharpe and HRP for each family. Equal Weight is the transparent reference. Minimum Variance uses the trailing covariance matrix to reduce estimated portfolio variance. Risk Parity spreads estimated risk contribution. Maximum Sharpe uses trailing

mean and covariance estimates to seek the largest excess return per unit of volatility. These quadratic methods descend from the return–variance trade-off formalised by Markowitz (1952).

HRP uses correlation distance, single-linkage clustering and recursive variance bisection. The method groups assets that moved similarly before distributing weight between clusters; it does not require the covariance inversion used by quadratic optimisation (López de Prado, 2016). Product constraints then project all methods onto the same long-only feasible set. Each equity position is capped at 10%, each cryptoasset at 25%, and the combined fund's crypto sleeve at 30%. The caps matter: they stop an unstable estimated mean or covariance from turning a fund into an unconstrained single-asset bet.

2.3 Timing, turnover and costs

Each monthly target uses a fixed trailing window ending before the rebalance date: 252 observations for Equity and Combined, and 365 for Crypto. The target is set on the first observed date of each month. Equity and Combined first go live on 4 January 2021; Crypto first goes live on 1 January 2021. Returns through 31 December 2020 therefore estimate the first targets and never appear as live fund performance. A temporal-leakage test changes future inputs and confirms that earlier weights and signals remain unchanged.

The performance path subtracts 10 basis points per unit of turnover. Initial deployment counts as 100% turnover; later turnover equals one half of the sum of absolute differences between the new target and the pre-trade drifted weights. A 50% rebalance therefore costs 0.05% of portfolio value under the model. The Sharpe ratio uses a 0% annual risk-free rate, which the brief permits. The app's separate 0.12% annual product fee is a product setting: an A\$10,000 balance implies A\$12 a year, but that fee is not deducted from the historical fund metrics.

The core net return identity is $r_{p,t} = w'_{t-1}r_t - 0.001 \times \text{turnover}_t$. The full configuration and sample definitions appear in Table 1. Growth paths in Figure 2 start at one dollar after costs; Figure 4 reports the combined-fund drawdowns, and Figure 5 shows how the five combined targets move through sectors and crypto over time.

3. Out-of-sample fund evidence

3.1 No method wins every family

No single method leads the equity family across every measure. Equity Equal Weight has the strongest equity return (12.6%) and Sharpe ratio (0.815). Equity Minimum Variance cuts volatility to 12.5% and maximum drawdown to 15.7%, but its return falls to 5.67%. Equity Maximum Sharpe returns 9.83% with a 0.647 Sharpe, below the equal-weight reference. Its average monthly rebalance turnover is 27.1%, compared with 5.43% for Equal Weight, and its latest effective asset count is 10.7 rather than 50. In this sample, the historical return estimate produces greater concentration without a better realised Sharpe.

Crypto Minimum Variance produces the largest annualised return in the menu, 69.5%, and the largest Sharpe ratio, 1.072. That result is not a low-risk outcome in an ordinary investor sense: annualised volatility is 76.9% and the fund loses 72.9% from its prior peak. The other crypto funds draw down between 78.1% and 82.7%. The method comparison therefore says less about a dependable return hierarchy than about how unusual the 2021–2023 crypto path is. Three live years are too short to treat a 69.5% compound rate as a forecast.

HRP behaves like a lower-risk alternative, not a return winner. Equity HRP posts 13.7% volatility and a 17.4% drawdown, but its 8.17% return trails Equity Equal Weight. Combined HRP records 14.0% volatility and a 17.9% drawdown, with 9.61% return. The low-risk behaviour is plausible given its cluster allocation, yet Minimum Variance still records lower volatility and a shallower drawdown in both families. HRP remains useful as a comparison because its clustering-based construction produces a different concentration pattern from the other methods; this sample does not establish superiority.

3.2 The combined family gives the clearest product comparison

Combined Risk Parity provides the strongest combined-fund Sharpe ratio, 0.883. It returns 13.9% a year with 16.2% volatility and a 19.5% maximum drawdown. Combined Equal Weight records the largest return, 15.0%, but its volatility is 21.6% and its drawdown reaches 27.9%. The return difference between the two funds is

1.17 percentage points a year; the drawdown difference is 8.41 points. A user choosing between them is choosing a historical risk profile, not simply accepting a lower number in exchange for nothing.

Combined Minimum Variance sits at the defensive end: 5.67% return, 12.5% volatility and a 15.8% drawdown. Combined Maximum Sharpe almost matches Equal Weight's return at 14.8% and improves drawdown to 19.4%, but it averages 27.7% turnover at each monthly rebalance and holds an effective 10.9 assets at the latest target. Risk Parity averages only 6.46% turnover and retains an effective 51.1 assets. The risk-adjusted result therefore comes with a broader and more stable allocation.

The risk–return scatter in Figure 3 keeps the crypto scale visible rather than compressing its risk into a verbal warning. The complete values in Table 2 use compound annual return, annualised daily volatility, a 0% risk-free rate and after-trading-cost returns. These labels matter because “net” does not include the 0.12% product fee.

3.3 Faster retraining raises turnover

The frequency experiment holds the combined universe, window, constraints and cost rate fixed for Risk Parity and Maximum Sharpe. Daily Maximum Sharpe reaches a 0.897 Sharpe versus 0.776 monthly, but annualised turnover rises from 333% to 1,412%. Under the 10-basis-point model, the ending-growth cost drag rises from 1.51 cents to 6.90 cents per initial dollar. The daily path is therefore more active than the phrase “daily rebalance” first suggests: it trades the equivalent of the portfolio more than fourteen times a year.

Monthly Risk Parity records a slightly higher Sharpe than daily rebalancing (0.883 versus 0.872), while requiring only 77.8% annualised turnover compared with 216% under daily rebalancing. Faster estimation is not a free accuracy upgrade; its value depends on method instability and trading. Figure 8 and Table 4 report this experiment. Monthly remains the only primary product schedule and the routine build no longer recomputes the faster paths.

4. Finance sentiment and fusion

4.1 Headline tone is kept separate from evidence quality

VADER supplies an inspectable rule-based sentiment score rather than a trained black box. Its original lexicon and language rules were designed for short social text (Hutto and Gilbert, 2014). Headlines are scored in their preserved textual form: the pipeline does not strip casing, punctuation or stopwords because VADER's rules use capitalisation, punctuation, intensifiers and negation when calculating valence. Finance headlines still expose predictable lexical gaps. Words such as “beat”, “upgrade”, “default” and “sell-off” can carry a domain meaning that differs from ordinary language. Ten blinded AI reviewers scored 34 candidate families on the VADER valence scale; the unweighted mean became the provisional score. I reviewed these candidate finance families, retained 33 unchanged and excluded “layoff” because its sentiment depends strongly on context. The final lexicon therefore contains 75 token variants.

The finance lexicon changes 9,481 of 146,836 headline compound scores and changes 5,721 positive/neutral/negative labels at the ± 0.05 threshold. The neutral share falls from 49.57% to 46.67%. Lower neutrality is not accuracy evidence. The audit retains the plain score, finance score, changed-label flag, token family and headline text so a reviewer can trace each correction.

Headlines first average within ticker-day. Every sector-day then equal-weights its five constituent tickers; a ticker-day with no headline receives zero. The rule stops a heavily covered company from receiving a larger portfolio signal merely because it has more articles. It also makes zero ambiguous unless coverage is shown beside tone, because zero can mean balanced language or absent news. A separate headline-level validation review marked five cases as ambiguous and reversed the expected direction of one case where sentence context changed the meaning of the isolated finance term.

Coverage confidence addresses that ambiguity. Breadth is the share of a sector's five companies with news, while headline-share HHI measures concentration. The score is $\text{breadth} \times (1 - \text{HHI}) / 0.8$, clipped to zero through one; no-news days equal zero. Materials has the lowest mean confidence, 0.392, and Consumer the highest, 0.823. The 0.431 gap means two similar sentiment values can rest on materially different evidence. Confidence says nothing about whether VADER classified the words correctly.

The ten sector paths in Figure 6 use a 21-observed-day mean for readability, while trading uses the separately stored prior-day value. Figure 7 plots coverage on the same sector structure. The matched layout makes thin evidence visible without treating it as a probability that the sentiment score is correct.

4.2 Two visible 2022 moves survive inspection, not without caveats

Utilities reaches a 21-day rolling score of 0.235 on 15 February 2022. The window contains 176 mapped headlines, 121 positive and five negative, with mean breadth of 83.8% and confidence of 72.5%. Earnings beats, outlooks, renewable investment plans and dividends explain the positive language. Plain VADER produces 0.215, so the finance extension adds only 0.020. Cross-ticker repetition accounts for 12.5% of mapped rows; retaining one date-title instance lowers the mean but does not remove the direction. I interpret the spike as credible evidence of positive headline tone, not as a measured change in expected utility-sector returns.

Technology's late-2022 line is often described as negative, but it remains above zero. Its rolling low is 0.051 on 25 October. The underlying window contains 571 mapped rows and broad coverage; weaker semiconductor demand, AMD's revenue warning, Intel slowdown reports, export restrictions, inflation and rate fears lower the score. Repeated titles are 24.0% of mapped rows, and several rhetorically bullish titles receive false negative VADER scores. The decline has an economic news basis, while its exact size is overstated by repetition and language errors. I left the primary index unchanged to avoid correcting the model after observing conspicuous events.

4.3 The sentiment tilt is a measured negative result

The fusion test starts from Equity Minimum Variance. On each monthly rebalance, the previous observed sector-day finance score is multiplied by its same-day coverage confidence. Cross-sectional sector scores are standardised, clipped at ± 2 and applied through $\exp(0.20 \times \text{score})$ before the 10% stock cap is restored. The one-day lag means a headline aligned to Monday can first affect Tuesday's decision; it can never alter Monday's weight.

The primary coverage-aware tilt lowers annualised return from 5.67% to 5.01% and Sharpe from 0.504 to 0.453. Volatility is almost unchanged at 12.5%. Maximum drawdown improves from 15.72% to 14.95%, a 0.77 percentage-point reduction, while average rebalance turnover rises from 14.5% to 16.6%. Table 3 and Figure 9 show the before-and-after evidence on identical dates and costs.

Sensitivity does not rescue the return claim. Tilt strengths of 0.10, 0.20 and 0.40 produce Sharpe ratios of 0.481, 0.453 and 0.401, all below the base 0.504. Stronger tilts make drawdown progressively shallower but sacrifice more return. I interpret this pattern as a small defensive reallocation, not a profitable forecast. The app therefore presents sentiment as evidence to inspect, and the tilt is excluded from the 15 investable menu funds because it does not improve return or Sharpe.

5. Investor journey and implementation

The app organises the research files around five investor tasks. Overview explains the product, process, families and methods. Compare funds starts with Selected funds and Benchmark, then treats family and method as filters rather than chart inputs. A user can compare up to five monthly funds through growth, return–volatility and an aligned evidence table. Same-family Equal Weight is the controlled default; SPY and ONEQ adjusted-price ETF series provide optional S&P 500 and Nasdaq Composite proxies on exact common dates.

Fund details opens one dated fact sheet. It places return, volatility, Sharpe and drawdown in a two-by-two grid, then shows benchmark growth, aligned drawdown, latest simulated targets, sector allocation, allocation through time and the largest target changes. I use “latest simulated targets” rather than “current holdings” because the final targets are dated 2023 observations. Downloads expose the return history and complete weight table.

Allocation lab lets the user divide 100% across selected funds with one segmented slider. It computes a monthly reset-and-drift historical mix, correlation, holdings overlap and look-through sector or crypto exposure. Equal allocation is the default test of the user's weighting choice; SPY and ONEQ are optional

market context. The fixed 0.12% product fee is shown in dollars rather than offered as a user setting. News signal then separates finance tone, coverage confidence, the plain-VADER check and the negative fusion result.

Precomputation keeps interaction fast because the deployed path loads committed CSV files only. It does not download the raw course bundle, run an optimiser or score VADER during interaction. A schema gate checks the required columns before rendering. Phase 8 records 76 canonical artifacts in a timestamp-free SHA-256 manifest; two builds match byte for byte. The full code suite contains 143 tests; the sentiment tests, Ruff checks, monthly build and artifact schema gates pass. A real Streamlit server returns HTTP 200. These checks verify reproducible outputs and app availability without recomputing the analysis during interaction.

6. Critical reflection and recommendations

6.1 Findings supported by the evidence

The walk-forward records provide an audit trail: every target has a prior-only window, an information cutoff, a rebalance date, a cost and a next-period return. Equal Weight prevents the optimiser comparison from becoming self-referential. Within the combined family, Risk Parity records the highest Sharpe with low turnover and broad effective holdings. HRP provides a different clustering rule and behaves defensively, even though it does not beat Risk Parity or Minimum Variance on their strongest metrics.

Coverage confidence helps distinguish similar sentiment scores that are supported by different amounts and distributions of news. It does not repair VADER or predict returns; it indicates whether a sector score represents broad, evenly distributed news. The event audits show why this separation matters: Utilities has broad support for its positive window, while Tech's direction survives even though repetition and phrasing distort the magnitude.

6.2 What did not work, and what remains unknown

Estimated optimisation does not consistently beat the simple reference. Equity Equal Weight leads its family on return and Sharpe. Combined Maximum Sharpe concentrates and trades more without beating Combined Risk Parity on Sharpe or Equal Weight on return. These are realised comparisons over three live years, not statistical proof that the methods are equal or that one will persist. The sample covers a sharp crypto boom and bust, inflation, rate tightening and a technology drawdown; a different regime may reverse the ranking.

The trading-cost model is transparent but thin. Ten basis points per unit turnover does not model bid-ask spreads by asset, market impact, capacity, taxes, custody, fund administration or crypto venue risk. I use a 0% risk-free rate to keep comparisons consistent, although it overstates excess returns when cash yields are positive. SPY and ONEQ are adjusted-price ETF proxies rather than official index series, and comparisons involving crypto use only shared US trading dates.

The sentiment extension has the clearest unresolved validity risk. VADER scores only titles, repeated broad-market stories can enter several ticker mappings, and 34 finance-family scores originate from AI review averages. The event audit finds both useful corrections and obvious false readings. The one-day lag prevents timing leakage, but it does not make headline tone causal. The fusion result itself is negative and may reflect weak predictive content, a blunt sector mapping, a poor tilt shape or a sample too short to distinguish those explanations.

App tests establish calculation and rendering behaviour, not customer understanding. A fresh-user comprehension session and logged-out public deployment test remain open. The prototype also lacks onboarding questions about loss tolerance, liquidity needs and investment horizon. It should not be presented as a regulated product or a substitute for personal advice.

6.3 Three directions for further investigation

These recommendations identify reasonable directions for further investigation. They are not product prescriptions because the evidence covers only the 2021–2023 historical simulation.

1. Freeze the current system and evaluate genuinely new data. Given the short evaluation period, a reasonable next step would be to freeze the 15 fund definitions, approved 75-token lexicon, cost assumptions

and signal rules before observing another period. Running the unchanged system on newly available data would test whether return, drawdown, turnover and allocation patterns persist. The test should predeclare one primary measure, such as net Sharpe, with drawdown and turnover guardrails rather than select whichever result looks favourable.

2. Investigate headline relevance before changing sentiment scores again. The completed manual review marked five cases ambiguous and reversed one because the full sentence changed the isolated finance word's meaning. A reasonable extension would classify headlines as company-specific, sector-wide or market-wide, identify repeated stories mapped to several tickers and then freeze those rules for a new holdout test. Until that evidence exists, the sentiment output is best interpreted as an inspectable panel rather than another fund in the menu.

3. Investigate estimation stability and market regimes together. The changing allocations and different method rankings suggest that a longer study could compare the static funds with shrinkage-based estimates and a simple regime-aware allocation. Two or three regimes could be defined in advance from lagged market trend, volatility, drawdown, cross-asset correlation and geopolitical-news intensity. Regime labels and switching rules should be fixed before the holdout and judged after costs using return, drawdown, turnover, concentration and weight stability. Geopolitical news would measure risk intensity, not provide a directional forecast.

7. Conclusion

Overall, no allocation method dominates across asset families or performance measures. Stockist Funds presents 15 monthly funds with prior-only weights, dated fact sheets, after-trading-cost paths, a standalone finance-sentiment index and a leakage-safe fusion experiment. Combined Risk Parity gives the strongest combined Sharpe; Equal Weight gives the strongest combined return; Minimum Variance gives the shallowest combined drawdown. Crypto produces the highest returns and the largest losses.

The unsuccessful sentiment tilt remains informative: inspectable news analytics do not automatically produce an investable edge. The app keeps that distinction visible through coverage, lags, benchmarks, holdings and plain-language limits. The evidence therefore supports a cautious next step: preserve the frozen systematic evidence, investigate headline relevance and regime sensitivity on new data, and test user comprehension before making claims beyond a 2021–2023 historical simulation.

References

- Australian Securities and Investments Commission (2026), “What is a managed fund”, Moneysmart, updated 1 July 2026. <https://moneysmart.gov.au/managed-funds-and-etfs/what-is-a-managed-fund>
- Hutto, C.J. and Gilbert, E. (2014), “VADER: A parsimonious rule-based model for sentiment analysis of social media text”, Proceedings of the International AAAI Conference on Web and Social Media, 8(1), pp. 216–225. <https://doi.org/10.1609/icwsm.v8i1.14550>
- López de Prado, M. (2016), “Building diversified portfolios that outperform out-of-sample”, The Journal of Portfolio Management, 42(4), pp. 59–69. <https://doi.org/10.3905/jpm.2016.42.4.059>
- Markowitz, H. (1952), “Portfolio selection”, The Journal of Finance, 7(1), pp. 77–91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>

Appendix A. Evidence tables

The tables report two to three significant digits from the machine-readable artifacts. Percentages are annualised where labelled. All fund results cover the family-specific live period in 2021–2023 and include the 10-basis-point turnover cost assumption but exclude the 0.12% product fee.

Table 1. Primary model design. Windows end before each first-of-month target. Equity and Combined use 252-day annualisation; Crypto uses 365. Source: model_configuration.csv.

Design choice	Primary specification
Asset universe	50 equities; 10 cryptoassets; Combined uses all 60
Portfolio methods	Equal Weight; Minimum Variance; Risk Parity; Maximum Sharpe; HRP
Trailing windows	Equity 252; Crypto 365; Combined 252 observations
Primary schedule	First observed date of each month after a complete window
First live dates	Equity/Combined 4 Jan 2021; Crypto 1 Jan 2021
Long-only caps	Equity 10%; cryptoasset 25%; Combined crypto sleeve 30%
Risk-free rate	0% annually
Trading cost	10 bp × turnover; 5 bp and 25 bp diagnostic sensitivities
Product fee	0.12% annually in app illustration; not deducted from fund metrics
Sample end	Equity/Combined 29 Dec 2023; Crypto 31 Dec 2023

Table 2. Out-of-sample performance of the 15 monthly funds. Return is CAGR; volatility is annualised daily standard deviation; Sharpe uses a 0% risk-free rate; drawdown is the largest fall from a prior after-trading-cost wealth peak. Source: report_performance_table.csv.

Family	Method	Return	Volatility	Sharpe	Max drawdown	Avg turnover	Effective assets
Equity	Equal Weight	12.6%	16.1%	0.815	-20.3%	5.4%	50.0
Equity	Hierarchical Risk Parity	8.2%	13.7%	0.642	-17.4%	14.7%	32.9
Equity	Maximum Sharpe	9.8%	16.6%	0.647	-19.2%	27.1%	10.7
Equity	Minimum Variance	5.7%	12.5%	0.504	-15.7%	14.5%	13.2
Equity	Risk Parity	9.8%	14.5%	0.718	-18.5%	5.9%	44.6
Crypto	Equal Weight	40.4%	81.9%	0.828	-81.6%	8.6%	10.0
Crypto	Hierarchical Risk Parity	42.4%	77.0%	0.848	-78.1%	13.0%	7.6
Crypto	Maximum Sharpe	14.8%	76.3%	0.566	-82.7%	22.5%	4.5
Crypto	Minimum Variance	69.5%	76.9%	1.072	-72.9%	15.9%	4.9
Crypto	Risk Parity	44.1%	79.9%	0.861	-79.9%	9.0%	9.4
Combined	Equal Weight	15.0%	21.6%	0.757	-27.9%	6.5%	60.0
Combined	Hierarchical Risk Parity	9.6%	14.0%	0.727	-17.9%	13.1%	34.6
Combined	Maximum Sharpe	14.8%	20.5%	0.776	-19.4%	27.7%	10.9
Combined	Minimum Variance	5.7%	12.5%	0.504	-15.8%	14.6%	13.4
Combined	Risk Parity	13.9%	16.2%	0.883	-19.5%	6.5%	51.1

Table 3. Equity Minimum Variance before and after sentiment tilts. Every path uses the same 2021–2023 dates, monthly targets, universe, constraints and 10-basis-point cost assumption. Delta columns compare with the base fund. Source: fusion_performance_comparison.csv.

Variant	Tilt	Return	Volatility	Sharpe	Max drawdown	Avg turnover	Δ Sharpe
Base Equity Minimum Variance	0.00	5.7%	12.5%	0.504	-15.7%	14.5%	+0.000
Plain-VADER tilt	0.20	4.8%	12.5%	0.439	-15.3%	16.4%	-0.065
Finance-VADER tilt	0.20	5.0%	12.5%	0.457	-15.1%	16.6%	-0.047
Coverage-aware finance tilt	0.20	5.0%	12.5%	0.453	-15.0%	16.6%	-0.050

Sentiment audit: 146,836 headlines scored; 9,481 compound scores changed; neutral share 49.57% plain versus 46.67% finance-adjusted. Mean coverage confidence ranges from Materials (0.39) to Consumer (0.82).

Table 4. Frozen rebalance-frequency diagnostics for the two selected combined methods at 10 basis points per unit turnover. Faster schedules are research comparisons, not product funds. Source: rebalance_frequency_metrics.csv.

Method	Schedule	Return	Sharpe	Max drawdown	Annual turnover	Growth cost drag
Maximum Sharpe	Monthly	14.8%	0.776	-19.4%	333%	\$0.015
Maximum Sharpe	Every 10 days	16.0%	0.842	-19.2%	472%	\$0.022
Maximum Sharpe	Every 5 days	15.7%	0.829	-19.7%	663%	\$0.031
Maximum Sharpe	Daily	17.1%	0.897	-18.5%	1412%	\$0.069
Risk Parity	Monthly	13.9%	0.883	-19.5%	78%	\$0.003
Risk Parity	Every 10 days	13.6%	0.875	-20.0%	98%	\$0.004
Risk Parity	Every 5 days	13.3%	0.865	-20.1%	121%	\$0.005
Risk Parity	Daily	13.4%	0.872	-20.2%	216%	\$0.009

Appendix B. Report figures

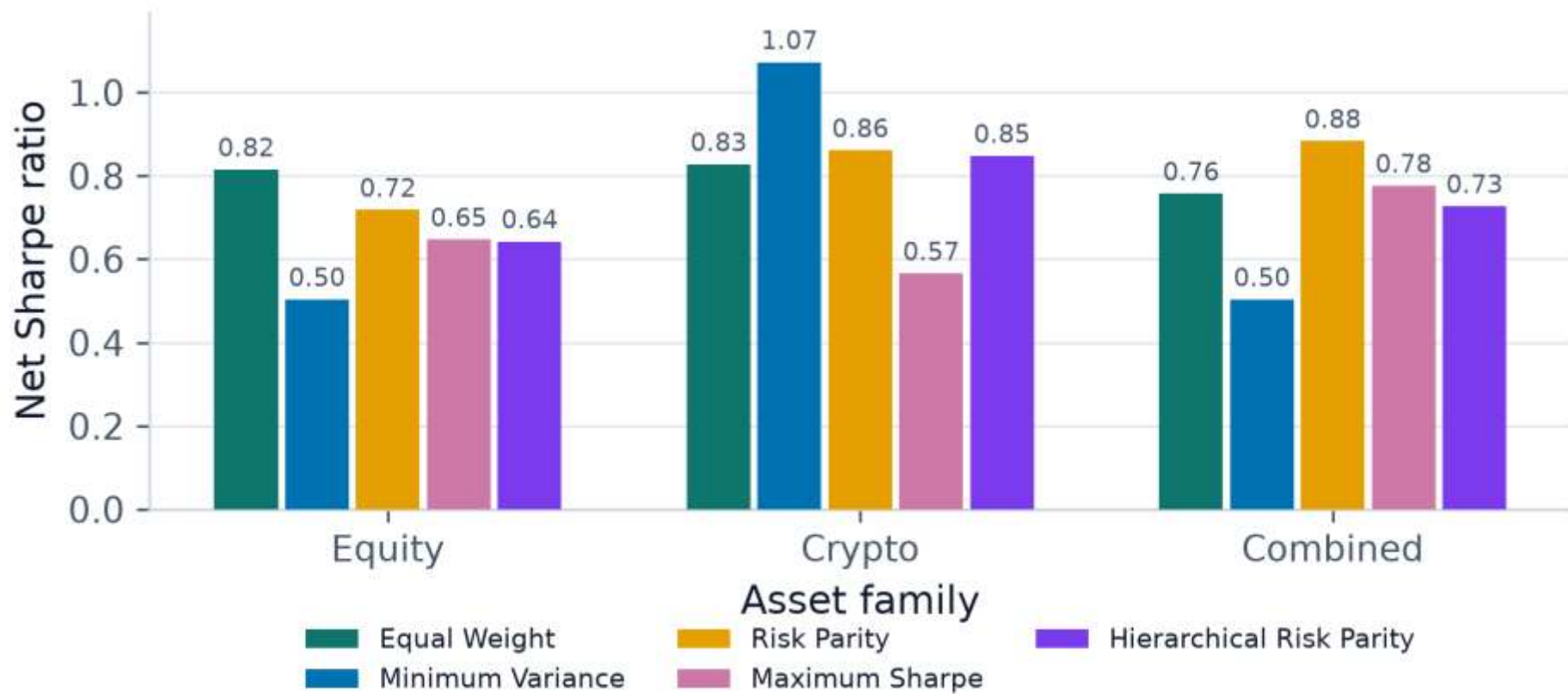


Figure 1. Out-of-sample Sharpe ratio by family and method. Bars compare the 15 monthly funds after the 10-basis-point turnover-cost assumption. Sharpe uses a 0% annual risk-free rate. Equity and Combined use 252-day annualisation; Crypto uses 365. Sample: 2021–2023. Source: course data and Stockist Funds calculations.

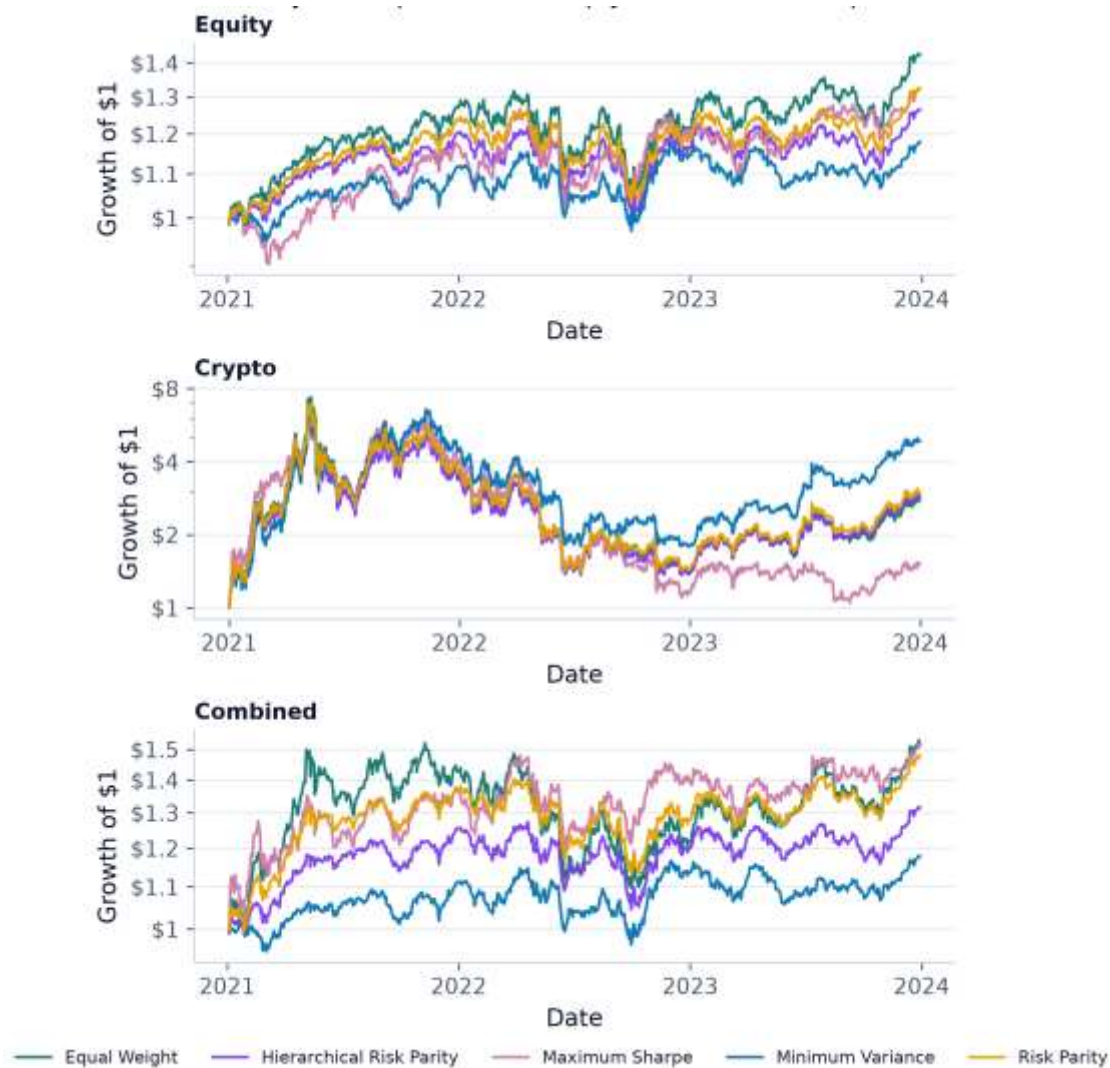


Figure 2. Growth of one dollar across the 15 monthly funds. Each panel starts at \$1 and uses a logarithmic scale so the wider crypto range remains readable. Paths use prior-only windows and include 10 basis points per unit turnover. Sample: 2021–2023. Source: course data and Stockist Funds calculations.

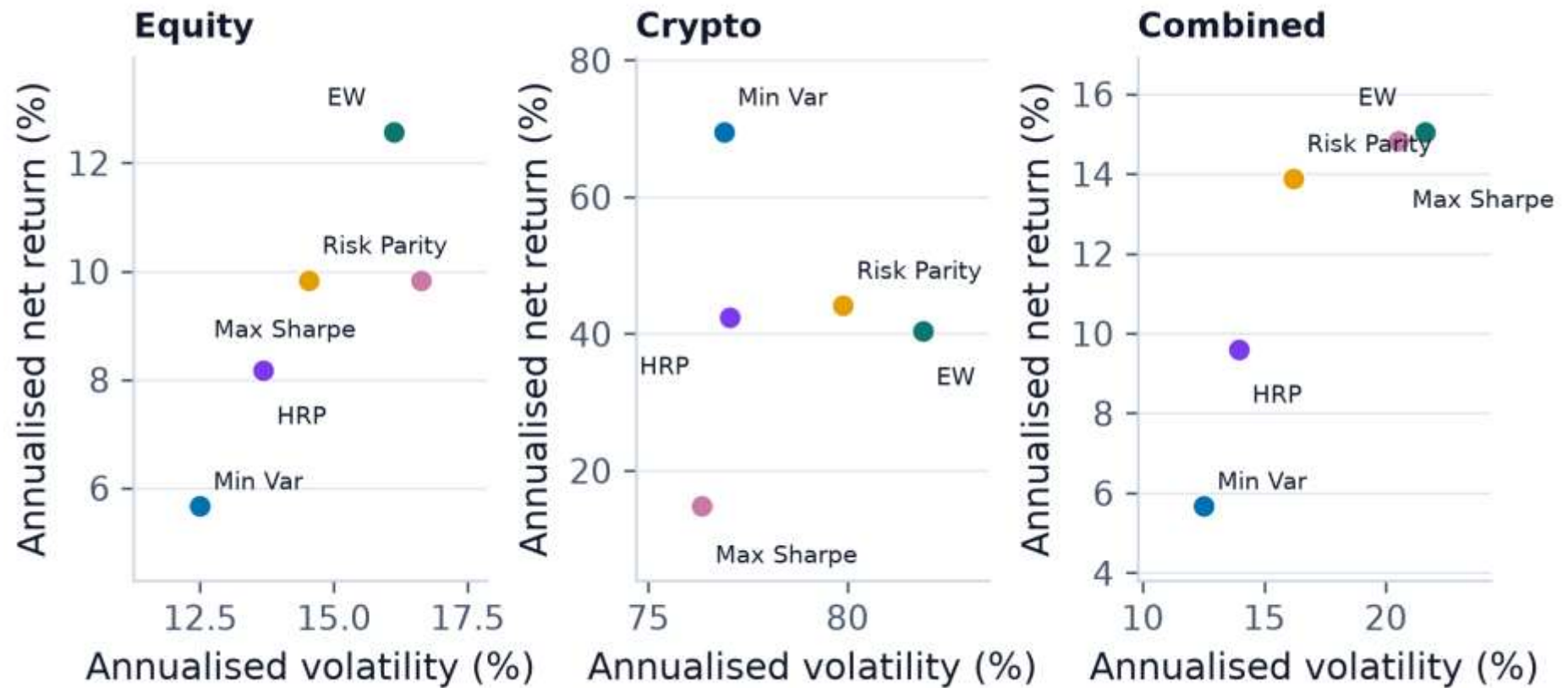


Figure 3. Annualised return and volatility across monthly funds. Each point is one family-method fund. Return is compound annual growth and volatility is annualised daily standard deviation after the trading-cost model. Sample: 2021–2023. Source: course data and Stockist Funds calculations.

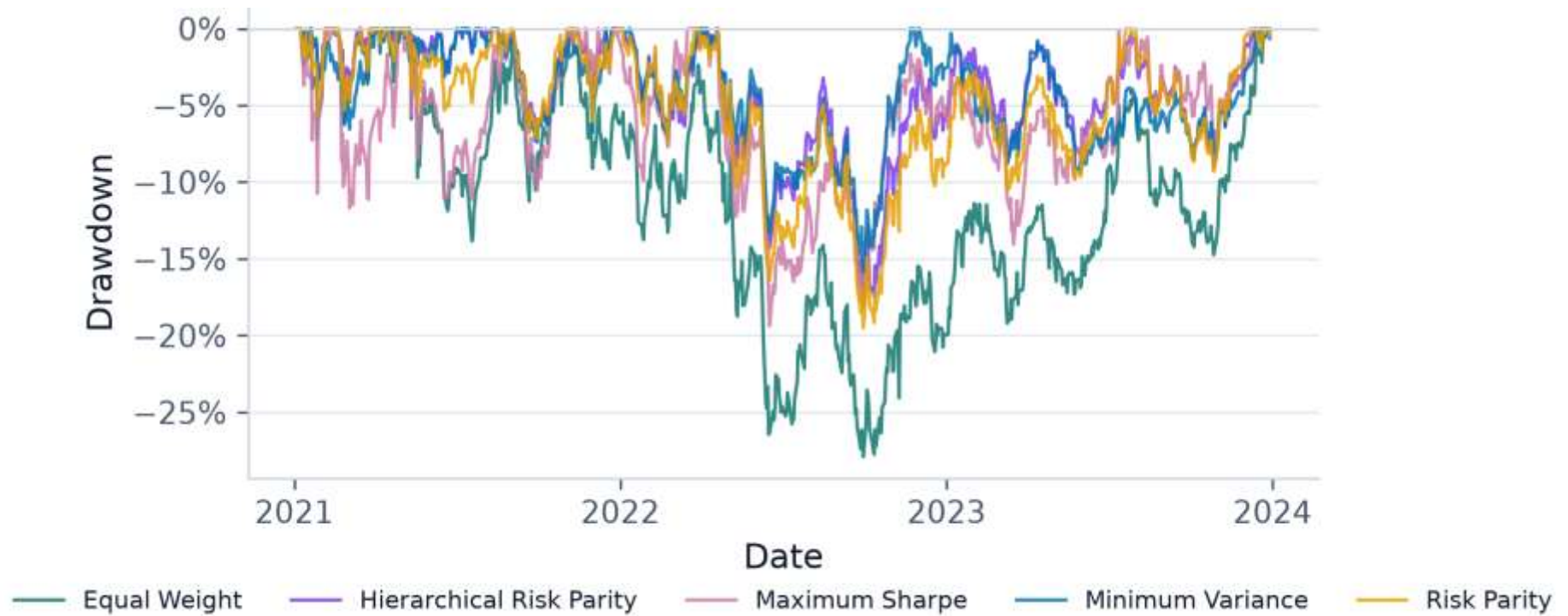


Figure 4. Drawdowns of the five combined monthly funds. Drawdown is the percentage decline from each fund's prior after-trading-cost wealth peak. Lower lines indicate deeper historical loss. Sample: 4 January 2021 to 29 December 2023. Source: course data and Stockist Funds calculations.

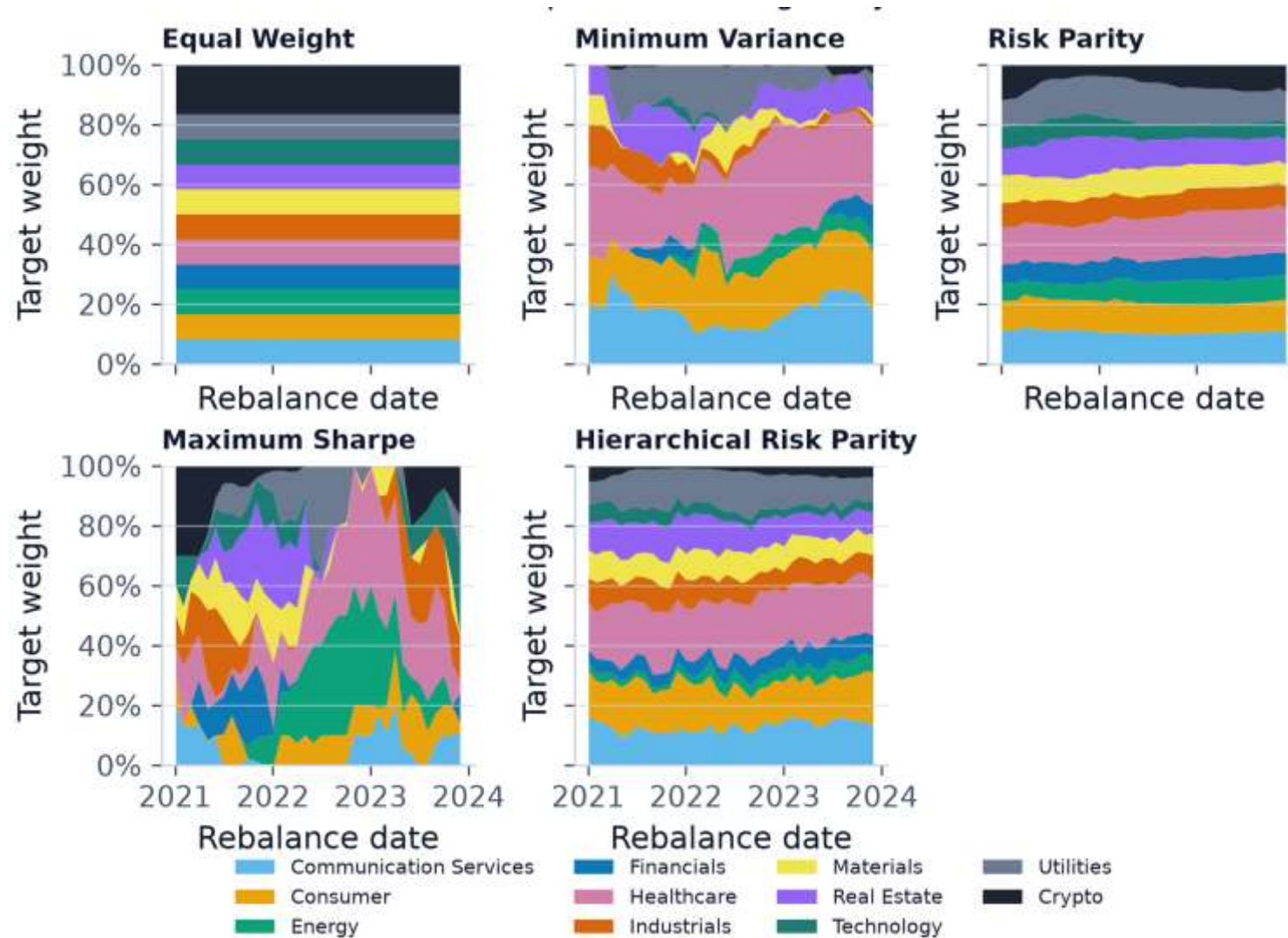


Figure 5. Monthly target allocation history for the five combined funds. Each band is an equity sector's target share; all cryptoassets form one Crypto band. Bands total 100% at every monthly target. Sample: 2021–2023. Source: course data and Stockist Funds calculations.

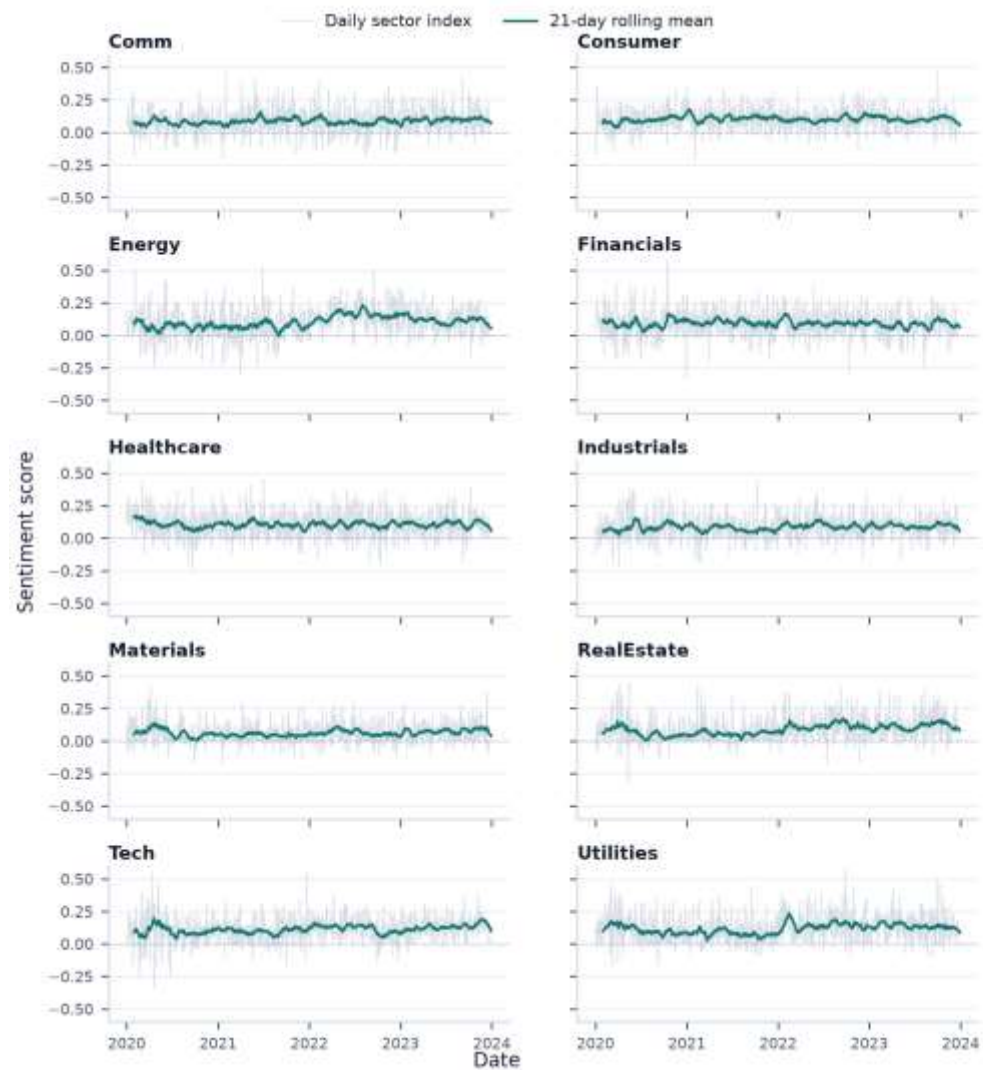


Figure 6. Finance-adjusted VADER sentiment across equity sectors. The index averages headlines within ticker-day, assigns zero to no-news ticker-days, then equal-weights five tickers per sector. Solid lines are 21-observed-day means; faint lines are daily values. Trading uses a separate one-day lag. Sample: 2 January 2020 to 29 December 2023. Units: VADER compound score on $[-1, 1]$. Source: course data and Stockist Funds calculations.

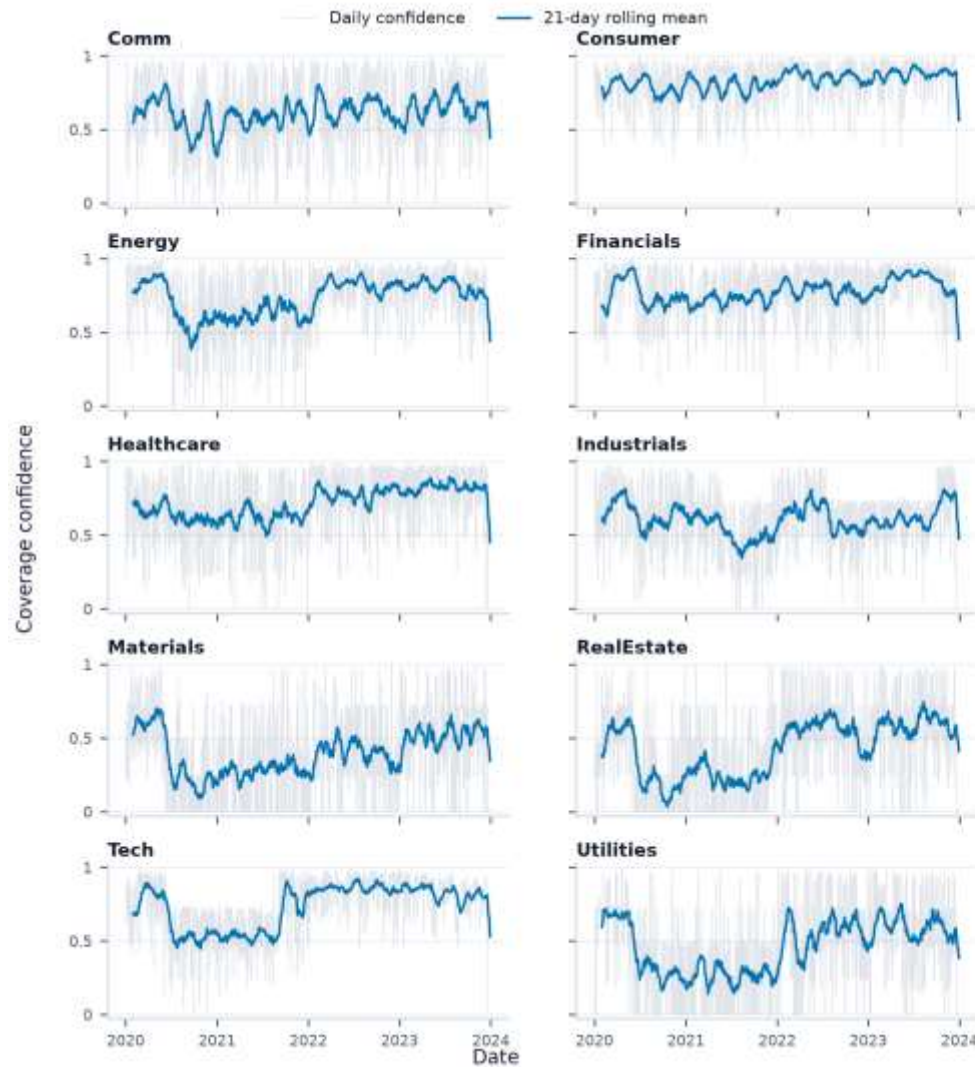


Figure 7. Coverage confidence for the ten sector sentiment indices. Confidence combines covered-company breadth and headline concentration, scaled to [0, 1]. Zero denotes no usable news; a high score means broad, more even coverage, not correct sentiment classification. Sample: 2 January 2020 to 29 December 2023. Source: course data and Stockist Funds calculations.



Figure 8. Turnover and Sharpe in the frozen frequency experiment. The experiment fixes method inputs and constraints, then varies monthly, every-10-day, every-5-day and daily targets for Combined Risk Parity and Maximum Sharpe. Faster paths are diagnostics. Sample: 4 January 2021 to 29 December 2023. Units: annualised turnover and after-trading-cost Sharpe at 10 basis points. Source: course data and Stockist Funds calculations.



Figure 9. Coverage-aware sentiment fusion versus Equity Minimum Variance. Both paths use identical monthly dates, eligible stocks, constraints and 10-basis-point costs. The augmented path applies a fixed 0.20 exponential tilt to prior-day finance sentiment multiplied by coverage confidence. Sample: 4 January 2021 to 29 December 2023. Source: course data and Stockist Funds calculations.

Appendix C. Reproducibility and AI-use disclosure

The routine build runs ``python scripts/run_part_b.py``. It regenerates the 15 monthly funds and core sentiment/fusion evidence, validates the frozen frequency comparison, checks 76 canonical artifacts and writes ``results/tables/artifact_manifest.csv``. ``python scripts/check_reproducibility.py`` runs the monthly build twice and compares relative paths, file dimensions, byte sizes and SHA-256 hashes. The deployed app reads only committed results files.

AI assisted with code generation, tests, method prototypes, interface revisions, the finance-lexicon scoring panels, event audits and the report draft. I retained responsibility for the final design and analytical decisions, including the product name, monthly primary schedule, constraints, 0.12% fee, five fund methods, app hierarchy and the decision to report negative results. I rejected or changed several AI suggestions, including the fixed negative score for “layoff”, a dark app theme, confusing evidence labels, editable product fees, customer-facing proprietary method details and routine recomputation of faster schedules. Numbered records under ``ai/`` document the prompts, errors, corrections and open review tasks.